Fiscal Year:	FY 2015	**Task Last Updated:**	FY 02/17/2016
PI Name:	Clark, Noel A. Ph.D.		
Project Title:	Observation and Analysis of Smectic Islands in Space (OASIS)		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	COMPLEX FLUIDS/SOFT MATTER--Complex Fluids		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Boulder	**State:**	CO
Zip Code:	80309-0001	**Congressional District:**	2
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	NOT AVAILABLE
Start Date:	09/01/2013	**End Date:**	08/31/2017
No. of Post Docs:	1	**No. of PhD Degrees:**	3
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	6
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA GRC
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Flight Program:	ISS		
Flight Assignment:	ISS		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Maclennan, Joseph Ph.D. (University of Colorado) Glaser, Matthew Ph.D. (University of Colorado) Stannarius, Ralf Ph.D. (Magdeburg University, Germany)		
Grant/Contract No.:	NNX13AQ81G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Detailed tests of theories of hydrodynamic flow, of relaxation of hydrodynamic perturbations, and of hydrodynamic interactions in 2D will be made. Freely suspended bubbles in microgravity, without islands, convection, and sedimentation represent nearly ideal, physically and chemically homogeneous 2D fluid systems for the precision study of 2D hydrodynamics. The effects of introducing islands or droplets will be studied, both as controllable inclusions that modify the flow and as markers of flow. A second goal will be to study the behavior of collective systems of 1D layer step interfaces on 2D bubble surfaces, including the equilibrium spatial organization and interaction of islands, and the nonequilibrium coarsening dynamics of island emulsions. In addition to yielding information about a number of relatively weak physical effects (thickness-dependent surface tension and line tension, disjoining pressure, etc.), we anticipate that this will clarify the effects of dimensionality in coarsening dynamics (e.g., on dynamic scaling behavior). Study interactions between islands and droplets in regimes where they are very weak, in the smectic A phase and at high temperature in the smectic C phase. The first thermocapillary experiments on homogeneous two dimensional fluids. The proposed microgravity experiments present a unique opportunity to explore thermocapillarity, the translational symmetry of the films and the absence of convection mitigating anomalous effects to the maximum extent possible, enabling detailed studies of the thermocapillarity of 2D fluids. Study the dependence of surface tension and line tension on film thickness and Burgers vector. Ground-based experiments indicate that equilibrium and nonequilibrium island behavior should be sensitive to this dependence, enabling critical tests of extant theoretical predictions. Probing the effects of a spontaneously broken symmetry in the 2D film surface (the appearance of 2D polar, XY-like ordering and accompanying electrostatic polarization) on the interaction of islands, exploring the stability of topologically stabilized emulsions of 1D interfaces in 2D. See also: http://spaceflightsystems.grc.nasa.gov/
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Liquid crystals can flow like a liquid, but contain molecules that are arranged in a specific pattern, just like a crystal. Greater understanding of the physics behind these structures could lead to improved liquid crystal display devices, including devices with improved color contrast and response times. It could also advance research in high-speed electro-optic devices used to control a light beam.
Task Progress:	Three-dimensional (3D) coarsening dynamics in emulsions, foams, and other non-equilibrium systems is an important and relatively well-studied problem. Bubbles of smectic liquid crystal, investigated in the OASIS experiments on the International Space Station (ISS), offer a well-characterized, homogeneous platform for the study of both equilibrium and non-equilibrium behavior of collective systems of 1D interfaces in 2D. There have been very few studies of such phenomena in fluids. We have already made the observations of the coarsening of island emulsions on the 2D surface of bubbles of smectic liquid crystal created on ISS. Air jets are used to create extensional flow in an initially homogeneously thin bubble film. This draws excess material from the bubble meniscus (which is confined to a small region near the needle used to blow the bubble) onto the bubble surface. The air flow also breaks this thicker material up into small, disk-like inclusions called islands that then form a 2D emulsion on the bubble surface. On Earth, such island emulsions either leak off of flat films at the meniscus edge, or sediment to the lowest point of a bubble and coalesce. However, in microgravity, coarsening of the emulsion can be observed free of such effects. Early observations on smectic A liquid crystal bubbles 15 mm in diameter confirm that an emulsion of small islands evolves to a significantly coarser ensemble of islands within about one hour. Much of the coarsening appears to be the result of pairwise island coalescence events. Further investigations will be carried out to determine how much Ostwald ripening might also be occurring and to characterize the dependence of coarsening on dimensionality. We also have observed what looks like the two-dimensional equivalent of the Plateau-Rayleigh instability in a thin bubble of smectic A liquid crystal (the 8CB mixture). In this experiment, we began by creating a thin bubble as usual. The temperature was raised locally to about 35 degrees using a heated needle located near the "north pole" of the bubble. A gentle stream of air applied across the top of the bubble (from one of the airjets) caused flow of the bubble material, which eventually entrained residual material originally located near the inflation needle, extending it into a continuous ribbon of thicker material along the flow direction that was swept around and around the bubble by the airjet. This steady-state motion of the ribbon could be continued indefinitely. However, when the heater needle temperature was increased to 45 degrees, a dramatic transition in the shape of the liquid ribbon occurred, with it breaking up into a chain of small disks (the two-dimensional equivalent of droplets) which then continue to flow around the film. It is proposed that this increase in temperature lowers the line tension (the two-dimensional equivalent of surface tension) of the ribbon sufficiently that it becomes unstable to small shape perturbations and pinches off into islands, much the same way that a water jet behaves in three dimensions.
Bibliography Type:	Description: (Last Updated: 12/04/2024)
Articles in Peer-reviewed Journals	Meienberg K, Malinina T, Nguyen Z, Park CS, Glaser MA, Clark NA, Maclennan JE. "Nanoparticle aggregation and fractal growth in fluid smectic membranes." Molecular Crystals and Liquid Crystals. 2015:611(1):14-20. http://dx.doi.org/10.1080/15421406.2015.1027990 , Jul-2015
Articles in Peer-reviewed Journals	Qi Z, Nguyen ZH, Park CS, Glaser MA, Maclennan JE, Clark NA, Kuriabova T, Powers TR. "Mutual diffusion of inclusions in freely suspended smectic liquid crystal films." Phys Rev Lett. 2014 Sep 19;113(12):128304. http://dx.doi.org/10.1103/PhysRevLett.113.128304 ; PubMed PMID: 25279649 , Sep-2014
Articles in Peer-reviewed Journals	Hu N, Shao R, Zhu C, Shen Y, Park C, Korblova E, Guerra C, Rego JA, Hexemer A, Clark NA, Walba DM. "Ferroelectric and antiferroelectric odd–even behavior in a tricarbosilane-terminated liquid crystal homologous series." Chemical Science. 2014 May 1;5:1869-74. http://dx.doi.org/10.1039/C3SC53353K , May-2014

Dissertations and Theses	Qi Z. (Zhiyuan Qi) "Hydrodynamics of Inclusions in Freely Suspended Liquid Crystal Films." Department of Physics, University of Colorado, May 2015. , May-2015